Marcus–de Oliveira Determinantal Conjecture

Counterexample Mining and an Exact $n = 4$ Boundary Certificate Through Eighth Order

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Computational research report

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Prepared for independent mathematical verification and extension.

Strongest result. At an $n = 4$ commuting/permutation point, every off-diagonal skew-Hermitian exponential direction is certified non-outward through order eight. Successive equality forces the odd coefficients of orders $3, 5, 7$ to vanish, while the even coefficients of orders $4, 6, 8$ have the inward sign. The first unresolved local coefficient is order nine.

Scope. This is an exact computer-assisted local theorem, conditional only on checking the supplied finite rational certificates and invariant-basis enumerations. It is not a proof of the global conjecture. No credible numerical counterexample was found in dimensions $4, 5, 6$. Novelty has not been independently established.

Companion programs, exact JSON certificates, raw searches, and tests are supplied separately in the reproducibility archive.

Contents

1 Executive summary

For normal complex matrices $A, B \in M_n(\mathbb{C})$, the Marcus–de Oliveira conjecture asserts

$$\det(A+B) \in \operatorname{conv}\left\{ \prod_{i=1}^{n} \big(\lambda_i(A) + \lambda_{\sigma(i)}(B)\big) : \sigma \in S_n \right\}.$$

After unitary conjugation and relabelling, the computational form is

$$A = \operatorname{diag}(a_1, \dots, a_n), \qquad B = U \operatorname{diag}(b_1, \dots, b_n) U^*.$$

The conjectured polygon depends only on the spectra a, b, whereas $\det(A+B)$ varies with U.

This project combined global counterexample search with exact local analysis. The global search found no violation in dimensions $4, 5, 6$, across generic, structured, rational, near-block, and near-permutation families. The local analysis produced the stronger mathematical output: an exact certificate that rules out every outward one-parameter mechanism through order eight at an $n = 4$ permutation unitary.

1.1 Precise local statement

Relabel B so the commuting point is the identity permutation, and write

$$z_\sigma = \prod_i (a_i + b_{\sigma(i)}), \qquad f(t) = \det\!\big(A + e^{tK} B e^{-tK}\big),$$

where K is off-diagonal and skew-Hermitian. Let $L : \mathbb{C} \to \mathbb{R}$ support the permutation polygon at z_{id}, and put

$$g(t) = L(f(t)) - L(z_{\mathrm{id}}).$$

The exact certificates establish the implication chain

$$\begin{aligned}
&g'(0) = 0, \qquad g''(0) \le 0,\\
&g''(0) = 0 \Longrightarrow g'''(0) = 0, \quad g^{(4)}(0) \le 0,\\
&g^{(4)}(0) = 0 \Longrightarrow g^{(5)}(0) = 0, \quad g^{(6)}(0) \le 0,\\
&g^{(6)}(0) = 0 \Longrightarrow g^{(7)}(0) = 0, \quad g^{(8)}(0) \le 0.
\end{aligned}$$

Thus, among Taylor terms of order at most eight, the first nonzero term cannot point outside the supporting line. This statement is local, one-parameter, and specific to $n = 4$; it does not rule out a global excursion or a mechanism whose first nonzero coefficient is order nine or higher.

1.2 Certificate inventory

Graph	Quartic dual rays	Order 6 rank	Order 7 rank	Order 8 rank
C_4	17	24	automatic	46
paw	11	21	10	39
$K_4 - e$	7	43	32 (34 raw)	101 (106 raw)
K_4	3	84 (90 raw)	automatic	175 on equality
Total	38			

Here “automatic” means that C_4 is bipartite and has no odd-degree gauge invariant, while the exact K_4 dual rays are invariant under permutation inversion and therefore annihilate every odd Taylor coefficient.

2 Status, normalization, and known safe regions

The conjecture remains open for $n \geq 4$. The current FormalConjectures Lean statement is marked research-open and contains a proof placeholder [1]. The search treated the following positive regions as controls or exclusions:

- $n \leq 3$;
- Hermitian pairs and the wider line/circle or essentially-Hermitian cases;
- scalar multiples of unitaries;
- sufficiently repeated spectra (including the known "two exceptional eigenvalues" regime);
- the proved $n = 4$ Householder case;
- direct sums of known pairs and normal-dilation constructions;
- Shitov's sufficient condition involving at most three nonreal eigenvalues in one spectrum [3];
- the singular $A + B$ variational result [4].

The recent survey of Bebiano and da Providência was used as the main status cross-check [2]. Kovačec's stronger transposition-edge polygon was implemented as a separate experimental target [5].

3 Minor-coordinate linearization

The central structural reduction uses Cauchy–Binet. For any unitary U,

$$\det(A + UBU^*) = \sum_{|I|=|J|} a_{I^c} b_J \, |\det U_{I,J}|^2. \tag{1}$$

For a permutation σ, the corresponding vertex has the parallel expansion

$$z_\sigma = \sum_{|I|=|J|} a_{I^c} b_J \, \mathbf{1}_{\{\sigma(I)=J\}}. \tag{2}$$

Consequently, a spectral support functional acts linearly on the finite vector of squared minors. At $n = 4$, there are 70 pairs (I, J) with $|I| = |J|$, but the 24 permutation-incidence columns have rank 14. All cone calculations are therefore performed in an exact 14-dimensional row basis.

This reduction is stronger than necessary: membership of the squared-minor vector in the permutation-incidence cone implies the desired statement for every spectral functional. Failure of that stronger cone inclusion would only produce an abstract minor-coordinate separator; it would not automatically produce spectra, because spectral functionals occupy a specially coupled, real-rank-at-most-two family at each fixed degree.

4 Second and third variations

Direct differentiation at the commuting point gives

$$f'(0) = 0, \qquad f''(0) = 2 \sum_{i<j} |K_{ij}|^2 \big(z_{(ij)} - z_{\mathrm{id}}\big). \tag{3}$$

If L supports the polygon at z_{id}, then every chord difference in (3) has nonpositive L-value, proving $g''(0) \leq 0$.

Define the active graph by the transpositions satisfying

$$L(z_{(ij)}) = L(z_{\mathrm{id}}).$$

If $g''(0) = 0$, then $K_{ij} = 0$ off this graph. Every cubic monomial is supported on at most three coordinate indices. Restricting to each such block and applying the known $n \le 3$ theorem for both signs of t forces its cubic coefficient to vanish. Summing the block contributions gives

$$g''(0) = 0 \implies g'''(0) = 0.$$

Equation (3) was also checked numerically in 360 random cases. Median relative finite-difference errors were 1.4×10^{-8}, 2.1×10^{-8}, and 2.3×10^{-8} for $n = 4, 5, 6$, respectively.

5 Exact quartic cone certificate

When the quadratic term vanishes, K is supported on the active graph. Up to relabelling, graphs with at most three edges reduce to the known $n \le 3$, Givens, disjoint-edge, or triangle-free mechanisms. The remaining graph types are C_4, the paw, $K_4 - e$, and K_4.

For each graph, the fourth-order squared-minor vector is projected modulo the identity and active-transposition columns. Every extreme ray of the dual cone is enumerated with exact `Fraction` arithmetic, then evaluated on a complete quartic invariant basis.

Graph	Quotient dim.	Candidate active sets	Dual rays	Patterns
C_4	9	75,582	17	10
paw	9	75,582	11	4
$K_4 - e$	8	31,824	7	5
K_4	7	12,376	3	3

For C_4, write $s_e = |K_e|^2$ and

$$P = \mathrm{Re}(K_{01}K_{12}K_{23}K_{30}), \qquad |P| \le \sqrt{s_0 s_1 s_2 s_3}.$$

Every one of its ten normalized patterns is nonnegative by monomial positivity or two-term AM–GM. A representative is

$$\tfrac{1}{2}(s_0 s_1 + s_2 s_3) - P \ge \sqrt{s_0 s_1 s_2 s_3} - P \ge 0.$$

For $K_4 - e$, with edge order $(01, 02, 03, 12, 13)$, the five forms are, up to positive scale,

$$s_2 s_3, \quad s_1 s_4, \quad s_1 s_4 + s_2 s_3 - 2P, \quad s_1 s_2 + s_3 s_4 \pm 2P.$$

For K_4, the three forms are the perfect-matching inequalities

$$s_0 s_5 + s_2 s_3 - 2P_{0123}, \quad s_0 s_5 + s_1 s_4 - 2P_{0132}, \quad s_1 s_4 + s_2 s_3 - 2P_{0213}.$$

All are exact AM–GM certificates. Duality yields $g^{(4)}(0) \le 0$.

6 Why the invariant interpolation is exhaustive

The higher-order scripts do not infer a polynomial from samples heuristically. They first establish a finite complete invariant space, and only then use exact samples to recover coordinates in that space.

Under diagonal unitary conjugation, a monomial in the entries of K and their conjugates can occur only if its directed multigraph is balanced at every vertex. Every balanced directed multigraph decomposes into directed simple cycles. On four vertices the relevant primitive cycle lengths are 2, 3, and 4. Entrywise conjugation forces real combinations. Therefore the partitions of the total degree determine complete templates:

Degree	Directed cycle types
5	$3+2$
6	$2+2+2,\ 4+2,\ 3+3$
7	$3+2+2,\ 3+4$
8	$2+2+2+2,\ 4+2+2,\ 3+3+2,\ 4+4$

The programs enumerate these templates, evaluate them on Gaussian-rational generators, select an exact full-rank subsystem by rational row reduction, solve the square system exactly, and validate on ten independent exact samples. Equality restrictions are then checked as exact Laurent-polynomial identities. No floating-point inference enters the theorem-producing path.

7 Orders five through eight

7.1 Fifth-order degeneracy

At degree five, the only new invariants have the form

$$s_e\ \mathrm{Re}(K_{ij}K_{jk}K_{ki}).$$

The paw and K_4-e bases have dimensions 4 and 10. On every paw ray, the fifth form is zero or is killed by the corresponding quartic monomial equality. For K_4-e, put

$$A = z_1\overline{z}_2, \qquad B = z_3\overline{z}_4.$$

The only nonzero cases accompany $|A \pm B|^2$, and substitution of $A = \mp B$ annihilates the fifth form. Odd order is automatic for C_4 and K_4 for the reasons stated in the executive summary.

7.2 Sixth-order nonnegativity

Complete invariant ranks are 21, 43, 24, and 84 for the paw, K_4-e, C_4, and K_4. On paw equality branches the residual is zero or a positive cubic monomial. For K_4-e, the nontrivial regular strata reduce to

$$\frac{s_1s_2}{36}\left[s_0(\rho+\rho^{-1}) + 2\,\mathrm{Re}(w^2)\right] \quad \text{or} \quad \frac{s_1s_2}{9}\left[s_0(\rho+\rho^{-1}) - 2\,\mathrm{Re}(w^2)\right],$$

where $\rho = |z_3/z_1|^2$ and $w = z_0(z_3/z_1)/|z_3/z_1|$. These are nonnegative because $\rho+\rho^{-1} \geq 2$ and $|\mathrm{Re}(w^2)| \leq s_0$.

The ratio parameterization assumes nonzero denominators. The exact script separately enumerates its four singular coordinate components. Two have zero sixth form; the other two retain positive residuals $s_0s_2s_3/36$ and $s_0s_1s_4/36$, or the same expressions scaled to $1/9$. This endpoint split is mathematically essential.

All C_4 equality substitutions, and all three K_4 matching substitutions, make the sixth form identically zero.

7.3 Seventh-order vanishing

Only the paw and K_4-e can carry degree-seven invariants. Their exact ranks are 10 and 32. On every joint quartic/sixth equality component, all 18 dual-ray forms vanish as Laurent-polynomial identities. The first still-possible term is therefore even.

7.4 Eighth-order sums of squares

For the paw, ten rays vanish. The surviving ray is

$$\frac{s_2}{144}\left(s_3(s_0^2+s_1^2)+2\,\mathrm{Re}(T_{012}^2)\right) = \frac{s_2}{144}\left(s_3(s_0-s_1)^2+4(\mathrm{Re}\,T_{012})^2\right)\geq 0.$$

For K_4-e, all perfect-square and phase-endpoint strata give zero. The monomial branches reduce to positive monomials or a two-term modulus square, certified by $|X|^2=D_1D_2$.

For C_4, most reductions vanish. The four survivors are Laurent squares; a representative is

$$\frac{s_1s_2}{144s_0^2}(s_0-s_1)^2(s_0-s_2)^2\geq 0.$$

The full K_4 degree-eight raw basis has dimension 294. On one perfect-matching equality variety it restricts to rank 175. Exact interpolation gives

$$\frac{|\overline{z}_0z_1^2+z_0z_3^2|^2|\overline{z}_5z_2^2+z_1^2z_5|^2}{144|z_1|^4}\geq 0. \tag{4}$$

The transitive S_4-action on perfect matchings carries (4) to the other two dual rays.

8 A useful false lead: the singular eighth-order endpoint

An intermediate reduction appeared to produce an outward eighth-order form on four K_4-e rays. One example was

$$-\frac{s_0s_2s_3(s_0+s_2+s_3)}{360}.$$

The associated abstract minor-coordinate facet could even be realized to machine precision by spectra, so this was a credible constructional lead rather than random hull noise.

Direct determinant evaluation nevertheless moved inward. The resolution was that a Laurent ratio used in the sixth-order formula had cancelled a denominator at a coordinate endpoint. On that endpoint, the sixth form is actually the strictly positive residual $s_0s_2s_3/36$. Imposing true sixth-order equality sets at least one of those three edge energies to zero, which also annihilates the negative eighth polynomial.

This episode is preserved as part of the certificate audit trail. It shows why every equality ideal must be stratified before division and why direct spectral realization is a valuable independent check even during exact algebraic work.

9 Global computational search

The principal separating objective was

$$\delta(a,b,U,\theta)=\frac{\mathrm{Re}(e^{-i\theta}\det(A+UBU^*))-\max_\sigma \mathrm{Re}(e^{-i\theta}z_\sigma)}{\max\left\{|\det(A+UBU^*)|,(|S_n|^{-1}\sum_\sigma|z_\sigma|^2)^{1/2}\right\}}.$$

A strictly positive value is a counterexample certificate. The search used Haar QR, Hermitian exponential and Cayley coordinates, exactly unitary Gaussian-rational Cayley matrices, Fourier/Hadamard matrices, Householder reflections, Drury's counterexample unitary to the stronger minor-convexity conjecture [6], near-block families, and perturbations of permutation unitaries. Spectra included generic complex Gaussians, Gaussian integers, repeated values, zeros plus roots of unity, and jointly optimized continuous families.

n	Direct samples	Fixed-U opts.	Joint restarts	Near-perm.	Outside
4	56,000	8	16	12,000	0
5	34,000	6	14	12,000	0
6	15,500	4	10	12,000	0

At least 169,588 differential-evolution objective evaluations were used in the fixed-unitary stages, followed by local polishing. The closest generic robust candidates remained inside under a separate 120-digit verifier:

n	Closest-edge support gap	Max squared unitarity residual
4	$-2.9325830189 \times 10^{-13}$	1.22×10^{-30}
5	$-5.2075426206 \times 10^{-13}$	2.27×10^{-31}
6	$-7.8439324943 \times 10^{-14}$	2.66×10^{-31}

Three independent membership implementations were required to agree: a local monotone-chain hull, Qhull half-spaces, and a convex-weight linear program. An apparent positive Cayley result of 2.72×10^{-11} was rejected because its squared unitarity residual was 2.98×10^{-10}, larger than the claimed separation.

9.1 Structured-unitary controls

Family	n	Best normalized support gap
Drury stronger-conjecture unitary	4	-1.0031×10^{-3}
Uniform Householder	4	-8.2820×10^{-4}
Uniform Householder	5	-2.5229×10^{-3}
Fourier	4	-6.9327×10^{-4}
Fourier	5	-1.9184×10^{-4}
Fourier	6	-5.7520×10^{-4}
Rational Cayley	4	-4.7177×10^{-4}
Rational Cayley	5	-8.9896×10^{-3}
Rational Cayley	6	-7.5043×10^{-2}

The stronger minor-convexity counterexample did not transfer automatically. At fixed degree a spectral separator has coefficient matrix $\mathrm{Re}(cx_I y_J)$, of real rank at most two, while Drury's displayed degree-two facet has rank three. Other degrees are coupled by the same spectra.

10 Test of Kovačec's stronger polygon

The nonconvex transposition-edge construction in [5] was implemented by splitting all segment intersections, constructing the planar half-edge arrangement, and extracting the unique outer face.

- On each of Kovačec's two published $n = 4$ spectral examples, 20,000 Haar unitaries produced no point outside the stronger polygon and no MOC violation.
- For $n = 5$, 12 generic spectral pairs with 3,000 Haar unitaries each (36,000 total) produced no point outside the stronger polygon and no MOC violation. The arrangements contained roughly 12,700–18,000 nodes and 13,100–18,400 bounded faces.

This answers the preprint's request for $n \geq 5$ experiments, but remains double-precision evidence rather than a proof.

11 Independent verification protocol

The following order is recommended. A checker can stop as soon as a dependency fails.

V1. Run the complete test suite and confirm 11 tests pass.
V2. Regenerate the quartic sparse and dense JSON files. Check the incidence rank 14, quotient dimensions, exact dual-ray counts, and every AM–GM pattern.
V3. Regenerate order five, then order six. Confirm invariant ranks and the ten independent validation samples per sixth-order graph.
V4. Inspect the singular $K_4 - e$ components in `results_sixth_order_exact.json`; do not infer endpoint behavior from the regular ρ-formula alone.
V5. Regenerate order seven and verify every joint equality substitution is the zero Laurent polynomial.
V6. Regenerate order eight. Check the paw identity, the $K_4 - e$ endpoint refinements, all four nonzero C_4 Laurent squares, and the rank-175 K_4 representative with factorization (4).
V7. Separately rerun the direct determinant evaluation on stored candidates and require agreement of all three convex-membership methods.

The theorem-producing path uses Python's `Fraction` class and a custom Gaussian-rational type. The exact cone enumeration, interpolation, and Laurent substitution code contains no numerical tolerance.

12 Reproduction commands

From the directory containing the companion programs:

```
python -m unittest -v

python moc_quartic_cone.py \
  --output results_quartic_cone_exact.json
python moc_quartic_dense_cone.py \
  --output results_quartic_dense_cone_exact.json
python moc_fifth_order.py \
  --output results_fifth_order_exact.json
python moc_sixth_order.py \
  --output results_sixth_order_exact.json
python moc_seventh_order.py \
  --output results_seventh_order_exact.json
python moc_eighth_order.py \
  --output results_eighth_order_exact.json
```

The exact sparse cone enumeration takes roughly 15 seconds; the remaining graph and higher-order certificates take roughly two minutes in the supplied pure-Python implementation. The full regression suite completed 11 tests in 135.630 seconds in the archived environment.

Global-search commands and all explicit seeds are recorded in `REPORT.md` and the corresponding JSON files.

13 What remains

The strongest next local experiment is order nine on the eighth-order equality strata. No new odd-order work is needed for C_4 or K_4. The genuinely new cases are the paw and $K_4 - e$, and the degree-eight certificates should be used to prune their equality ideals before constructing a ninth-order basis.

Globally, the most informative complementary directions are:

- continue testing the sharper Kovačec polygon at $n = 5$ and $n = 6$;
- search for a way to realize an abstract minor-coordinate outward facet inside the coupled spectral rank-two family;
- study non-exponential unitary jets, or prove that diagonal gauge removes the apparent additional jet freedom;
- formalize the finite exact cone and polynomial certificates in Lean.

No counterexample exists in the present data. If one is found later, the ideal certificate remains compact: rational or algebraic spectra, an exactly unitary rational/algebraic matrix, a rational real-linear separator, exact normality and determinant checks, and finitely many strict separator inequalities.

A Program and certificate manifest

The companion archive contains the complete search pipeline. The following are the theorem-critical files; SHA-256 hashes identify the versions used for this report.

```
5c8cf63647f7e4ef5d4db38c911e4c3360da0b7f1e60399521a658bd3e0d587b  moc_quartic_cone.py
78baa3ddc7499659e62502c8d9b155093817589ff180580c0cb28283d4f4901d  moc_quartic_dense_cone.py
5aa91301037f19294aed972a8d14e7c75b0391802b77cb297afe7e0c88dd0722  moc_fifth_order.py
a91115cd4dd7292f59174489ad0487a138493aba4074fffd852514612272be32  moc_sixth_order.py
aba55bb38bdcda31d674c46c04edd5675e80dfeef389011aaa14c2c640df6463  moc_seventh_order.py
a8205e089cf35a1430fc2138e1049eeeac643f0fc2dd0b53971176952c41547f  moc_eighth_order.py

10e8a93ab8bb25d7723022839b6a59b1f461a656ff072c694208fe30aa2d02c2  results_quartic_cone_exact.json
8753046c887870276f9bdcb7ec1bca30999b0d0d52f18be4e888898e5e8d7a83  results_quartic_dense_cone_exact.json
6c51c7d8201c41f01d748036383fdea7ab2d2180ec676dc7b2e977e692edf25b  results_fifth_order_exact.json
64c7779b7eaf6dd6969c8f60b03d70327fbfaf796f0e23324ffb91709f7f0efa  results_sixth_order_exact.json
1ce6afff839181cd2faa582cf6cbdbd5d727d0cf1134fe1208063c491c84cdfb  results_seventh_order_exact.json
24eaa5bc49369f9367e151493b4827633a473232d9201b4a1c266d600dc73845  results_eighth_order_exact.json
```

The archive also includes the global search, high-precision verifier, stronger polygon implementation, raw numerical outputs, and all tests.

B Certificate dependency graph

Stage	Depends on
Quadratic	Direct differentiation and support inequalities
Cubic	Quadratic equality plus the known $n \leq 3$ result
Quartic	Exact rank-14 minor coordinates, graph census, 38 dual rays, AM–GM
Fifth	Complete degree-five invariant bases and quartic equality ideals
Sixth	Complete degree-six bases and exact singular/regular stratification
Seventh	Complete degree-seven bases and joint quartic/sixth equality
Eighth	Complete degree-eight bases, Laurent squares, and S_4 symmetry

References

[1] Google DeepMind, *FormalConjectures/Wikipedia/DeterminantalConjecture.lean*. https://github.com/google-deepmind/formal-conjectures/blob/main/FormalConjectures/Wikipedia/DeterminantalConjecture.lean

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